TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

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Abstract

Objective: Thyroid eye disease (TED) can accompany Graves disease (GD), an autoimmune cause of hyperthyroidism. We compared genetic evidence linking blood *TSHR* and *IGF1R* expression with susceptibility.

Methods: We screened 2,544 druggable genes using Mendelian randomization (MR). Disease outcomes were Biobank Japan (BBJ) GD, UK Biobank (UKB) hyperthyroidism and overlapping FinnGen GD, Graves ophthalmopathy (GO) and strict GO endpoints. Colocalization tested 24 genes for a shared causal variant. Tissue eQTL and orbital RNA-sequencing (four TED, one control) were supplementary.

Results: Thirteen genes passed BBJ discovery ($P < 0.05/2{,}544$). Higher genetically proxied *TSHR* expression was associated with lower GD odds in BBJ (odds ratio [OR] per approximately one standard deviation 0.12, 95% confidence interval [CI] 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$), concordantly across all five datasets. Shared-variant probability exceeded 0.80 in FinnGen GO and, prior-sensitively, BBJ and strict GO, but UKB (0.226) and FinnGen GD (0.264) favored distinct variants. *IGF1R* ORs exceeded 1 but reached $P < 0.05$ only in BBJ (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB ($P = 0.0117$); the most precise estimate, FinnGen GD, was OR 1.07 (95% CI 0.86–1.33). Single-variant tissue estimates added no independent evidence. Mean orbital *TSHR* and *IGF1R* transcript levels exceeded the control.

Conclusions: Association evidence was stronger for *TSHR* than *IGF1R*, but absent colocalization in UKB and FinnGen GD leaves expression-mediated causality unproven. Population-controlled data cannot isolate TED susceptibility within GD; weaker *IGF1R* evidence does not argue against IGF-1R–targeted treatment.

Keywords: Graves disease; thyroid eye disease; *TSHR*; *IGF1R*; Mendelian randomization; colocalization

Introduction

Thyroid eye disease (TED) can cause proptosis, diplopia and visual impairment in patients with Graves disease (GD) [1, 2]. Teprotumumab, which blocks the insulin-like growth factor 1 receptor (IGF-1R), improves clinical outcomes in active TED [3]. Its efficacy raises a related but distinct question: does inherited variation influencing *IGF1R* expression contribute to disease susceptibility in the same way as variation at *TSHR*, the established GD autoantigen and susceptibility locus?

TSHR and IGF-1R interact in experimental models of orbital disease [4, 5, 6]. However, treatment response, inherited gene regulation and gene expression in inflamed tissue address different biological questions. A receptor may be a useful treatment target without equally strong genetic evidence linking its expression to disease onset. Distinguishing these forms of evidence helps clinicians interpret genetic studies of therapeutic targets.

Mendelian randomization (MR) uses variants associated with an exposure to investigate its relationship with the disease phenotype being studied, termed the outcome [7]. Here, the exposure is blood gene expression and the outcomes are disease diagnoses, not responses to treatment. Because alleles are allocated at conception, MR is less prone than ordinary observational studies to confounding and reverse causation, but it supports causal interpretation only when its instrumental-variable assumptions hold. Colocalization asks whether expression and disease associations in a genomic region are consistent with the same causal variant [8]. Distinct nearby variants can appear connected because they are inherited together. Thus, an MR association may be highly significant even when the regional signals do not support a shared variant. MR reports odds ratios (ORs), confidence intervals (CIs) and *P* values; colocalization reports model-based posterior probabilities, which are not *P* values.

Previous genetic studies have already implicated *TSHR* in GD and Graves ophthalmopathy. Ji and colleagues integrated blood methylation, expression and protein associations with GD, reporting *TSHR* among the expression-associated genes also supported in a strict FinnGen ophthalmopathy dataset [9]. Other MR work has examined inflammatory proteins in ophthalmopathy [10], and recent integrative research has investigated *TSHR* across genetic and transcriptomic data in GD and ophthalmopathy [11]. These findings establish a basis for comparing clinically relevant receptors, but do not determine whether evidence for their blood expression is consistently supported across disease definitions under the same analysis framework. In particular, limited evidence for *IGF1R* could reflect imprecision or dependence on particular instruments, while a strong MR association for *TSHR* does not by itself establish that expression and disease share a causal variant. The remaining question is how robustly these two expression traits are associated with GD and related phenotypes, and how far those associations support biological interpretation.

To address this question, we compared genetically proxied blood expression of *TSHR* and *IGF1R* — expression predicted from inherited variants rather than measured directly — within a screen of genes encoding potentially druggable proteins, using *CTLA4* and other associated genes as context. We used Biobank Japan (BBJ) GD for discovery, compared the original findings with UK Biobank (UKB) hyperthyroidism and FinnGen ophthalmopathy, and subsequently added FinnGen GD and a stricter ophthalmopathy definition. We evaluated effect direction, precision and statistical support across these datasets, then examined dependence on individual instruments, allele frequencies and colocalization assumptions. A later fixed-panel tissue extension evaluated whether these patterns extended beyond the original blood instruments. Our aim was to characterize the consistency and limits of expression-related genetic evidence for the two receptors. The analyses distinguish GD from broader hyperthyroidism and population-controlled ophthalmopathy; they do not directly

estimate TED susceptibility among patients who already have GD or predict treatment response. A small in-house orbital RNA-sequencing series (four TED samples and one control) is reported only as descriptive context.

Methods

Study design and disease datasets

This two-sample MR study used public association summary statistics and followed STROBE-MR reporting guidance [12]. The exposure used whole-blood expression quantitative trait locus (eQTL) statistics from eQTLGen, comprising up to 31,684 participants, predominantly of European ancestry [13]. The original outcome hierarchy comprised BBJ GD, UKB self-reported hyperthyroidism or thyrotoxicosis, and FinnGen R12 Graves ophthalmopathy [14, 15, 16]. We subsequently added FinnGen R12 E4_GRAVES_STRICT and E4_GRAVES_OPHT_STRICT. The five datasets and their case and control counts are shown in Table 1. The UKB source did not establish an exclusively European analysis sample, so ancestry was not inferred from the reference panel used in its source study.

BBJ remained the discovery dataset. FinnGen GD provided a same-disease comparison, UKB a broader-phenotype comparison, and the two ophthalmopathy endpoints sensitivity analyses of a TED-enriched phenotype. The three FinnGen endpoints contain overlapping participants and were not treated as independent replications or combined by meta-analysis. Their controls were population non-cases for the respective endpoint, not GD patients without TED. The FinnGen extensions, expanded candidate assessment and instrument diagnostics were post hoc; no prospective registration was available.

Expression instruments and MR

From 4,462 potentially druggable genes [17], instruments were selected from significant blood cis-eQTLs ($P < 5\times10^{-8}$) within 1 Mb of the gene. PLINK clumping used a linkage disequilibrium (LD) threshold of $r^2 < 0.001$ within 10 Mb and 503 European participants from 1000 Genomes Phase 3 [18, 19]. This produced 6,135 single-nucleotide polymorphism (SNP)–gene records for 2,544 genes. The minimum F statistic was 29.7; the median variant-specific expression sample size was 28,092, with a range of 232–31,684. Exposure coefficients and standard errors were reconstructed from Z statistics, sample sizes and reference allele frequencies under unit expression variance [20]. Alleles were aligned with TwoSampleMR; ambiguous palindromic or unavailable variants were excluded. UKB linear-model coefficients and standard errors were divided by $\mu(1-\mu)$, where $\mu = 3{,}731/484{,}598$, to approximate a log-odds scale. BBJ and FinnGen coefficients already used log odds.

We used Wald ratios for single-instrument genes and multiplicative random-effects inverse-variance weighting (IVW), with correction for under-dispersion, for multiple instruments [21]. An instrumental variable (IV) is a genetic variant used to represent variation in expression. Results are reported as ORs per approximately one standard deviation higher blood expression, with 95% CIs and two-sided *P* values. This scale describes reconstructed expression effects, not a drug dose or an intervention. Primary discovery required $P < 0.05/2{,}544$ (1.965×10^{-5}), retaining all eligible genes in the denominator even when an outcome estimate was unavailable. The 13 BBJ discoveries were assessed in FinnGen GD using a focused threshold of 0.05/13 and concordant effect direction. A separate full FinnGen screen used 0.05/2,544 and was exploratory. Nominal $P < 0.05$ was reported descriptively and was not equated with discovery significance.

MR interpretation assumes that instruments predict expression, are not associated with exposure–disease confounders, and influence disease only through the modeled exposure. A small *P* value or high PP.H4 cannot by itself establish these assumptions.

Where applicable, sensitivity assessment included Cochran's Q, alternative MR estimators and omission of individual instruments [22, 23]. Previously completed alternative-estimator results (MR-Egger, weighted median and weighted mode) for the original three outcomes were retained (Supplementary Dataset, MR_estimators_original); they were not represented as new analyses of every endpoint. Post hoc leave-one-out results assessed dependence on individual variants, not independent replication. Single-instrument *TSHR* results could not undergo these omission or multi-instrument pleiotropy tests.

In a post hoc allele-direction audit, we re-extracted rs179252 and rs179247 from the original expression and disease files. Disease effects were aligned to the blood expression-increasing allele, and G-allele correlations were calculated in European and East Asian reference genotypes and independently checked with PLINK. This audit assessed allele alignment and consistency with published thymus findings, rather than thymic mediation (Supplementary Dataset, TSHR_allele_audit).

Colocalization and robustness assessment

We report MR associations as the main findings and use colocalization to assess their interpretation. In the original analysis, colocalization was reported alongside MR as a parallel evidence layer; the present emphasis was adopted during post hoc reappraisal and was not a prospectively registered hierarchy. Colocalization used complete available regional cis-eQTL statistics, including non-significant variants, matched to each disease dataset [8]. The expanded set included all 13 BBJ discoveries, *IGF1R*, nine additional FinnGen GD discoveries and *HLA-G*, which was selected because its MR significance changed after

frequency substitution: 24 genes in total. We used coloc.abf with disease as dataset 1 and expression as dataset 2. Posterior probability of hypothesis 4 (PP.H4) measures support for a shared causal variant under the fitted model. PP.H3 instead supports distinct variants. We used PP.H4 ≥ 0.80 as a descriptive strong-support threshold, with priors $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$, and repeated analyses at $p12 = 5\times10^{-6}$ and 10^{-6} [24]. All five hypotheses are defined in Supplementary Methods. Low PP.H4 alone does not establish no disease association, and H3 does not prove multiple independent signals.

We evaluated reference and eQTLGen cohort frequencies, distinguishing changes in effect scaling from changes in variant availability and harmonization. The 24-gene colocalization dataset contains 1,440 settings: 24 genes × five outcomes × four frequency/availability scenarios × three priors. For *TSHR*, we additionally examined narrower windows, a common five-outcome variant set, raw regional disease peaks and all available source cis-eQTLs. For newly evaluated candidates, we checked marginal signal coverage, shared instruments and variant aliases. *C4B* and *HLA-G* underwent instrument-level frequency, weighting, heterogeneity and omission checks. Full-precision results are supplied in one Supplementary Dataset workbook. The original BBJ–ophthalmopathy combined filter is retained as a historical descriptive comparison, not promoted to a prospectively specified validation rule or the primary conclusion of this reappraisal.

Tissue-specific extension

In a post hoc extension, we assessed both genes in a fixed panel of five GTEx v8 tissues: thyroid, cultured fibroblasts, subcutaneous adipose, visceral adipose and whole blood [25]. The extension plan was fixed before retrieving these tissue results, after the blood analysis was known. GTEx participants were predominantly of European ancestry, not exclusively European. Instruments required cis-eQTL $P < 5\times10^{-8}$, $F > 10$ and European-reference LD $r^2 < 0.001$. Ten planned gene–tissue comparisons against FinnGen GD formed the primary family

($P < 0.005$); 40 comparisons against the other four datasets formed a secondary family ($P < 0.00125$). Unestimable combinations remained in the correction denominators. Effects used the original slopes and standard errors for inverse-normal-transformed expression. Complete regional summaries were not obtained for this extension, so tissue colocalization was not performed; significant-only pairs were used solely for instrument selection. All planned combinations are reported in the Supplementary Dataset.

Orbital expression and implementation

An in-house orbital RNA-sequencing series of four patients with inactive TED and one control provided descriptive context only (Supplementary Methods). Technical replicates were combined by biological sample, and transcript abundance was displayed without differential-expression significance testing (Figure S1). These observations were not used to select genes or validate the genetic findings. Analyses used R 4.3.3, TwoSampleMR 0.7.4, coloc 5.2.3 and PLINK 1.9, with independent numerical verification in Python. No age- or sex-stratified patient analyses, individual-level multivariable regression, Steiger directionality test or multi-signal colocalization was performed.

Results

Discovery across druggable genes

Of the 2,544 instrumented genes, 2,234 were estimable in BBJ, 2,505 in UKB and 2,480 in each FinnGen endpoint. Thirteen genes passed the BBJ screen threshold ($P < 0.05/2,544$), including *TSHR* and *CTLA4* but not *IGF1R* (Figure 1). At this threshold, 14.6% and 35.6% of estimable BBJ genes had 80% power to detect ORs of 1.5 and 2.0, respectively (Supplementary Dataset, Detection_original).

TSHR and IGF1R across disease datasets

Higher genetically proxied blood *TSHR* expression was associated with lower odds of GD in BBJ (OR 0.12, 95% CI 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$). ORs were also below 1 for UKB hyperthyroidism and both ophthalmopathy endpoints, and all estimates except strict ophthalmopathy passed the screen threshold (Table 2; Figure 2). Each estimate used the single instrument rs179252. Its expression-increasing G allele had per-allele disease ORs of 0.77–0.82 across datasets and was correlated with rs179247-G (European r = 0.978; East Asian r = 0.996), which also increased blood *TSHR* expression. Allele-aligned recalculation reproduced the MR estimates (Supplementary Dataset, TSHR_allele_audit).

All five *IGF1R* ORs exceeded 1, and none passed the screen threshold. Nominal associations (uncorrected $P < 0.05$) were found only in BBJ GD (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB hyperthyroidism (OR 1.35, 95% CI 1.07–1.70; $P = 0.0117$). The most precise estimate, FinnGen GD, was OR 1.07 (95% CI 0.86–1.33; $P = 0.539$); this comparison had 80% power to detect an OR of 1.36 at $P < 0.05$ (Supplementary Methods). Weighted-median ORs were 1.56 ($P = 0.0288$) in BBJ, 1.42 ($P = 0.00244$) in UKB and 1.46 ($P = 0.124$) in FinnGen ophthalmopathy. In these three datasets, MR-Egger and weighted-mode estimates had $P \geq 0.32$, and MR-Egger intercept *P* values were 0.66, 0.56 and 0.43, respectively (Supplementary Dataset, MR_estimators_original).

Other associated genes

Four genes passed the screen threshold in both GD datasets: *TSHR*, *CTLA4*, *HLA-DQA2* and *TUBB* (Table 3). *CTLA4* was associated with lower odds, and *HLA-DQA2* and *TUBB*, both in the major histocompatibility complex (MHC) region, with higher odds. *CTLA4* and *TUBB* had direction-consistent nominal associations in all five datasets; *HLA-DQA2* did in four, but its two-instrument UKB estimate was non-significant and below 1 (Supplementary Dataset,

MR_primary). At the focused FinnGen GD threshold ($P < 0.05/13 = 0.00385$), seven of the 13 BBJ discoveries had direction-consistent associations: these four genes plus *C4A*, *TNFSF14* and *IFNGR1*. *HLA-A*, which had the smallest BBJ *P* value, had a non-significant FinnGen GD estimate.

The exploratory full FinnGen GD screen identified 13 genes, nine of them outside the BBJ discovery set (Table S1). Seven of the nine were instrumented in the extended MHC region, which was not excluded from the screen, and two pairs each shared one instrument.

Instrument and frequency sensitivity

Omitting rs13030124 left non-significant *CTLA4* estimates in UKB and all three FinnGen endpoints ($P = 0.323–0.972$; Figure S2). In BBJ and UKB, where *IGF1R* had four instruments, omitting each in turn gave $P > 0.05$ in two of the four estimates; omitting rs2654980 in UKB also lowered the OR from 1.35 to 1.16 (Figure S2). Among the BBJ discoveries, Cochran's Q test in FinnGen GD indicated heterogeneity among instruments for *C4A* ($P = 1.27\times10^{-12}$), *TUBB* ($P = 3.67\times10^{-5}$), *HLA-A* ($P = 0.0387$) and *CTLA4* (Table 2). Omitting rs149543464 (*TUBB*) or rs1269852 (*C4A*) raised the FinnGen GD *P* value to 0.0262 and 0.0471, respectively (Supplementary Dataset, LOO_new_fixed14).

With eQTLGen cohort allele frequencies, all 13 BBJ discoveries still passed the screen threshold, *TSHR P* values were unchanged and no *IGF1R P* value crossed 0.05 (Supplementary Dataset, MR_original_freq and MR_new_freq). Screen classification changed only in FinnGen GD: *C4B* no longer passed, and *HLA-G* newly passed; the IVW weight of rs150254595 in its estimate rose from 21.3% to 58.9% (Table S2; Supplementary Dataset, Instrument_weights). Omitting this variant left non-significant *HLA-G* estimates under either frequency source (Supplementary Dataset, LOO_frequency).

Colocalization

TSHR PP.H4 (probability of one shared causal variant) was 0.951 in BBJ, 0.986 in FinnGen ophthalmopathy and 0.891 in strict ophthalmopathy, but 0.226 in UKB and 0.264 in FinnGen GD, where PP.H3 (distinct variants) was highest (Table 2; Figures 2 and 3). The UKB and FinnGen GD values were unchanged in narrower regions and in a variant set common to all five datasets, and the matched variants included the strongest regional disease variant in each dataset (Table S3). Lowering the prior probability of a shared variant (p12) from 10^{-5} to 10^{-6} gave PP.H4 of 0.875 in FinnGen ophthalmopathy, 0.661 in BBJ and 0.451 in strict ophthalmopathy.

IGF1R PP.H4 ranged from 0.010 to 0.404 (UKB); in the other four datasets, PP.H2 (expression signal only) was highest (0.571–0.731). *CTLA4* PP.H4 was ≥ 0.80 in UKB and both ophthalmopathy endpoints but not in BBJ or FinnGen GD. None of the 24 genes reached PP.H4 ≥ 0.80 in FinnGen GD, and *HLA-DQA2* and *TUBB* had PP.H4 < 0.001 in every dataset. Apart from *TSHR* and *CTLA4*, only *TNFSF14* and *IFNGR1* reached PP.H4 ≥ 0.80 in any dataset, both in BBJ only (Figure 3).

Tissue-specific extension

Four of the ten gene–tissue combinations had a qualifying instrument, each a single SNP: *TSHR* in subcutaneous and visceral adipose and *IGF1R* in thyroid and cultured fibroblasts. Thus, 20 of the 50 planned comparisons were estimable and 30 were not (Table S4). All ten adipose *TSHR* ORs were below 1, and eight, including both primary FinnGen GD comparisons, passed the multiple-testing threshold of their test family. The adipose instruments, rs12101255 (subcutaneous) and rs74064796 (visceral), lie 50 bp apart in *TSHR* intron 1, about 15 kb from rs179252 (European r^2 0.46 for rs12101255), and both are blood *TSHR* eQTLs (eQTLGen $P = 4.1\times10^{-34}$ and 6.5×10^{-34}). With one instrument, each tissue *P* value equals that variant's disease-association z-test *P* value, so the adipose estimates re-

express the regional *TSHR* association. No thyroid or cultured-fibroblast *IGF1R* estimate reached $P < 0.05$, and directions were inconsistent across datasets.

Descriptive orbital expression

Mean orbital *TSHR*, *IGF1R* and *CTLA4* transcript levels in four patients with inactive TED exceeded those in the single control, whose sample came from a different fat depot; no significance testing was performed (Figure S1).

Discussion

Higher genetically proxied blood *TSHR* expression was associated with lower GD odds in BBJ and FinnGen GD, with concordant directions for UKB hyperthyroidism and both ophthalmopathy endpoints. All *IGF1R* ORs exceeded 1, but only the BBJ and UKB estimates reached uncorrected $P < 0.05$, none met the discovery threshold, and the most precise estimate lay close to the null. Under the primary prior, colocalization supported a shared variant for *TSHR* in BBJ and both ophthalmopathy endpoints but favored distinct variants in UKB and FinnGen GD. These results do not establish a difference in biological importance between the receptors, and statistical significance in one gene but not another is not a formal test of different effects.

Our findings extend an existing body of genetic evidence. Ji and colleagues had already linked blood *TSHR* expression to GD and strict ophthalmopathy in an earlier FinnGen release [9]; recent integrative work has also investigated *TSHR* in GD and ophthalmopathy [11]. We add a focused comparison with *IGF1R*, a GD comparison spanning BBJ and FinnGen, and an assessment of both genes alongside the wider druggable-gene screen. Because Ji and colleagues also used eQTLGen and FinnGen, agreement is not wholly independent replication. Different MR methods, FinnGen releases and colocalization settings also preclude interpreting differences between reports as direct contradictions.

Among the other associated genes, the multi-instrument FinnGen and UKB *CTLA4* estimates relied heavily on rs13030124; with that caveat, *CTLA4*'s direction-consistent associations across all five outcomes support its use as an autoimmune comparator [26]. *HLA-DQA2* was discordant only in UKB, where its two instruments disagreed, while FinnGen GD *TUBB* and *C4A* estimates each depended largely on one variant. The strongest BBJ association, *HLA-A*, did not persist in FinnGen GD, so the smallest discovery *P* value need not identify the most transferable association. Shared instruments, complex MHC linkage and, for *HLA-DQA2* and *TUBB*, absent shared-variant support further prevent counting every associated expression trait as an independent causal target.

The lower-odds direction for *TSHR* need not contradict the pathogenic role of receptor-stimulating antibodies. Blood transcript abundance, tissue-specific expression and receptor activation are different exposures, and genetic regulatory effects can differ across tissues [25]. Our audit linked the blood expression-increasing rs179252-G allele to rs179247-G, the GD-protective allele associated with greater thymic transcription in published studies [27, 28]. This directional consistency is compatible with a central-tolerance explanation, but the thymic step was not tested here, and neither linked variant is a proven causal regulator. The marked departure of the MR ORs from 1 reflects scaling: rs179252 shifts reconstructed blood expression by only about 0.11 SD, and its per-allele disease ORs were 0.80 (BBJ), 0.77 (UKB), 0.78 (FinnGen ophthalmopathy), 0.77 (FinnGen GD) and 0.82 (strict ophthalmopathy). Rescaling these moderate per-allele effects to one SD of expression yields the per-SD ORs, which describe lifelong inherited differences, not the effect of raising *TSHR* expression or of TSHR-directed treatment. Because the adipose instruments tag the same intron-1 region as rs179252 and are themselves blood eQTLs, their concordant estimates re-express the regional association; they cannot resolve the distinct-variant pattern in UKB and FinnGen GD or establish an adipose-mediated mechanism, and neither depot is orbital fat.

Colocalization does not replace the MR associations, but it constrains causal wording. Strong *TSHR* associations in UKB and FinnGen GD coexisted with distinct-variant regional patterns, and BBJ and strict ophthalmopathy support weakened under a lower shared-variant prior. Thus, association strength and shared-variant evidence were not interchangeable, as also seen for the comparator *CTLA4* in BBJ and FinnGen GD. UKB and FinnGen GD had the most cases of the five datasets; their distinct-variant *TSHR* patterns could reflect finer resolution of a complex regional architecture, but this was not tested. Ji and colleagues chose a shared-variant prior intended to allow weaker signals to colocalize and accepted PP.H4 above 0.50 [9], whereas our primary settings used $p12 = 10^{-5}$ and PP.H4 ≥ 0.80, so colocalization labels are not directly comparable between the studies. The regional checks excluded a simple missing-peak explanation for *TSHR* but did not establish the correct causal architecture.

Experimental and clinical studies provide context for relating TSHR-directed immunity to orbital disease. TSHR can undergo cleavage and release its extracellular A-subunit. Expression of this subunit in an adenoviral immunization model induced greater thyroid-stimulating autoimmunity than a minimally cleaving receptor [29]; plasmid immunization subsequently reproduced orbital adipogenesis, fibrosis and muscle pathology in mice [30]. Daser and colleagues further reported enhanced inflammatory and hyaluronan responses to TNFα in orbital fibroblasts from TSHR-immunized mice [31]. Graves ophthalmopathy has also been documented in primarily euthyroid and hypothyroid patients [32], supporting a distinction between thyroid functional status and orbital immune activity. These observations make the ophthalmopathy associations biologically plausible, but our instruments do not measure receptor cleavage or antibody activity. Because most patients with TED have GD and the ophthalmopathy endpoints used population controls, their MR estimates may largely reflect GD susceptibility rather than a TED-specific effect, and higher colocalization probabilities do not establish one.

For *IGF1R*, blood expression-related genetic evidence was limited: ORs above 1, two nominal associations, imprecise ophthalmopathy estimates and mostly weak regional disease signals do not establish a susceptibility effect. Some single-variant omissions removed nominal significance, although most left ORs close to the full estimates. The near-null FinnGen GD estimate argues against large, but not modest, blood-expression effects. Thyroid and cultured-fibroblast *IGF1R* estimates were non-significant and directions were inconsistent across datasets, but single instruments, wide intervals and absent orbital eQTLs leave tissue-specific effects open. Cultured fibroblasts are not orbital fibroblasts, and unestimable combinations are not negative findings. Weaker genetic evidence for *IGF1R* does not argue against IGF-1R blockade: teprotumumab efficacy rests on a randomized trial in active TED [3], and experimental TSHR–IGF-1R crosstalk [5] concerns receptor signaling rather than inherited susceptibility.

Frequency substitution preserved the principal findings and changed full-screen classifications only for exploratory *C4B* and *HLA-G*; like instrument omission, it reused the same data and is not independent replication. For research on TSHR and IGF-1R, blood expression-related associations are more consistently supported for *TSHR*, while its tissue mechanism and the extent of an inherited *IGF1R* contribution remain unresolved. Independent expression instruments and tissue-relevant regulatory data would help address these uncertainties. Comparing GD patients with and without TED, while accounting for follow-up and selection into the GD group, would address orbital susceptibility directly. The present results inform interpretation and follow-up research, but do not guide treatment selection.

Limitations

The principal limitations involve the exposure and phenotype definitions. Predominantly European blood eQTL effects were applied to BBJ, requiring expression associations and LD

patterns to transfer across ancestries. UKB self-reported hyperthyroidism is broader than GD, and an exclusively European source sample was not established. FinnGen's overlapping, population-controlled endpoints do not directly compare GD patients with and without TED; cohort differences and phenotype differences cannot be separated cleanly. Participant overlap with eQTLGen could not be quantified. Cohort-frequency replacement, using a release excluding Framingham participants, addresses only part of expression reconstruction and does not remove LD-reference limitations or independently validate rare-variant effects. The orbital series, from inactive TED, has a single control and no inferential role. The GTEx extension, also predominantly European, had four estimable combinations, each with one instrument. It reused the disease datasets and lacked tissue-specific regional colocalization.

The analytical limitations include post hoc selection, few instruments and a single-causal-variant colocalization model. *TSHR* has one retained European-reference instrument, precluding multi-instrument pleiotropy tests; several other genes have few variants or heterogeneous estimates. MHC analyses are particularly vulnerable to complex LD and correlated expression, and low PP.H4 can reflect model mismatch or limited signal coverage. Multi-signal colocalization [33], directionality testing and formal between-phenotype effect tests were not performed. Only 24 selected genes underwent colocalization; therefore its negative classifications cannot be generalized to the entire screen. Discovery and focused follow-up thresholds address specified families of tests, while the expanded scans and sensitivity analyses remain exploratory. At the discovery threshold, most BBJ genes lacked 80% power for an OR of 1.5, and neither a non-significant MR result nor a posterior below 0.80 establishes absence of an effect.

Conclusions

Genetically proxied blood *TSHR* expression showed strong, directionally consistent associations with lower odds of GD and related phenotypes, whereas *IGF1R* evidence was

weaker, with the most precise estimate near the null. Colocalization supported a shared *TSHR* variant in FinnGen ophthalmopathy and, depending on the prior, in BBJ and strict ophthalmopathy, but not in UKB or FinnGen GD, so these data establish association rather than expression-mediated causality. Weaker genetic evidence for *IGF1R* does not argue against the therapeutic relevance of IGF-1R blockade. The population-controlled ophthalmopathy analyses do not estimate TED susceptibility among patients with GD or predict treatment response.

Declarations

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Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval. This study used publicly available, de-identified summary statistics and an institutionally approved orbital tissue dataset. The in-house orbital transcriptomic component was approved by the Institutional Review Board of Pusan National University Hospital (approval number 2104-018-102) and was conducted in accordance with the Declaration of Helsinki.

Informed consent. Written informed consent was obtained from all individual participants included in the in-house orbital tissue study.

Data availability. Instruments, primary MR and colocalization results, and sensitivity results are supplied as Supplementary Dataset. Analysis code is available from the corresponding author on reasonable request. Public summary statistics analyzed in this study are available from their original repositories: blood *cis*-eQTL data from the eQTLGen Consortium, tissue eQTL data from GTEx v8, the Biobank Japan Graves disease and UK Biobank hyperthyroidism genome-wide association statistics through the GWAS Catalog, and the

FinnGen Release 12 GD and ophthalmopathy statistics from FinnGen. The in-house orbital RNA-seq data are available from the corresponding author on reasonable request, subject to institutional and ethical restrictions.

Author contributions. J.P. conceived and designed the study, performed the analyses, and drafted the manuscript. M.-S.K. contributed to data collection and interpretation. K.-H.S. and S.-W.Y. supervised the study and revised the manuscript.

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Figure legends

Figure 1. Study design and gene-level discovery in BBJ. (A) Analysis flow. Blood expression instruments from eQTLGen were tested against BBJ GD for discovery; UKB hyperthyroidism and FinnGen GO were the original comparison datasets, and FinnGen GD, FinnGen GO strict and the GTEx tissue extension were added post hoc. The three FinnGen endpoints overlap and are not independent replications. (B) Each grey point is one estimable gene (n = 2,234), placed at the GRCh37 position of its strongest selected expression instrument and plotted by its BBJ MR *P* value; points are genes, not single variants. Filled black points are the 13 genes that passed $P < 0.05/2{,}544$ (dashed line, $-\log_{10}P \approx 4.71$); the open circle is the focal gene *IGF1R*, which did not. The threshold applies to the BBJ discovery screen, not to later exploratory analyses. GTEx, Genotype-Tissue Expression; MR, Mendelian randomization; SNP, single-nucleotide polymorphism; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure 2. *TSHR*, *IGF1R* and *CTLA4* across five disease datasets. (A) MR odds ratios with 95% CIs per approximately 1 SD higher genetically predicted blood expression: Wald ratio for one instrument, multiplicative random-effects IVW for several. The axis is logarithmic and the dashed line is OR = 1. *TSHR* used one instrument, rs179252, in every dataset. Colours identify datasets. (B) Colocalization posterior probabilities for the same comparisons ($p1 = p2 = 10^{-4}$, $p12 = 10^{-5}$), stacked from zero as H4 (one shared variant), H3 (distinct variants), H2 (expression signal only) and H0/H1 (neither trait, or disease only). The dotted line marks PP.H4 = 0.80, the descriptive reporting threshold. Estimates, *P* values and posterior probabilities are given in Table 2; *P* values and posterior probabilities answer different questions. FinnGen GO controls include people with GD but no recorded eye disease. CI, confidence interval; IVW, inverse-variance weighted; MR, Mendelian

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Table 1. Data sources and their roles

Dataset / endpoint	Cases	Controls	Role
eQTLGen blood cis-eQTL	—	—	Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc same-disease comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy (TED-enriched) comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc ophthalmopathy definition sensitivity; Finnish
GTEx v8 tissue eQTL	—	—	Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4)
1000 Genomes Phase 3 European reference (n = 503)	—	—	LD clumping of instruments and reference allele frequencies

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	*P* value	Q-test *P*	PP.H2	PP.H3	PP.H4	PP.H4, p12 = 10^{-6}
TSHR	BBJ GD	1	0.12 (0.07–0.21)	1.09×10^{-14}*†	NA	<0.001	0.049	**0.951**	0.661
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	3.19×10^{-32}*†	NA	<0.001	0.736	0.264	0.035
TSHR	UKB hyperthyroidism	1	0.09 (0.06–0.14)	8.77×10^{-28}*†	NA	<0.001	0.774	0.226	0.028
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	2.82×10^{-7}*†	NA	<0.001	0.014	**0.986**	**0.875**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	9.02×10^{-5}*	NA	0.027	0.082	**0.891**	0.451
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	0.0212*	0.902	0.690	0.236	0.073	0.008
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.882	0.731	0.259	0.010	0.001
IGF1R	UKB hyperthyroidism	4	1.35 (1.07–1.70)	0.0117*	0.250	0.400	0.196	0.404	0.063
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.309	0.623	0.346	0.032	0.003
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.409	0.571	0.391	0.038	0.004
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	5.45×10^{-15}*†	NA	<0.001	0.799	0.201	0.025
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	2.15×10^{-6}*†	0.0275	<0.001	0.746	0.254	0.033
CTLA4	UKB hyperthyroidism	2	0.21 (0.08–0.57)	0.00220*	0.00131	<0.001	0.047	**0.953**	0.672
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	0.0118*	0.0486	<0.001	0.022	**0.978**	**0.816**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	0.00888*	0.0755	<0.001	0.021	**0.979**	**0.820**

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124, and three of the four have Q-test $P < 0.05$; in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$ unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; eQTL, expression quantitative trait locus; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

BBJ discovery gene	Chr	IVs, BBJ/FinnGen GD	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	FinnGen GD PP.H4
HLA-A	6 (MHC)	4/4	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	6 (MHC)	1/1	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	2	1/2	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	14	1/1	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	6 (MHC)	1/3	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	16 (16p11.2 cluster)	1/2	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	6 (MHC)	2/3	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	16 (16p11.2 cluster)	1/0	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	19	1/1	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	16 (16p11.2 cluster)	1/1	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	12	4/4	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	6 (MHC)	1/1	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	6	1/1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; MR, Mendelian randomization; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure 1

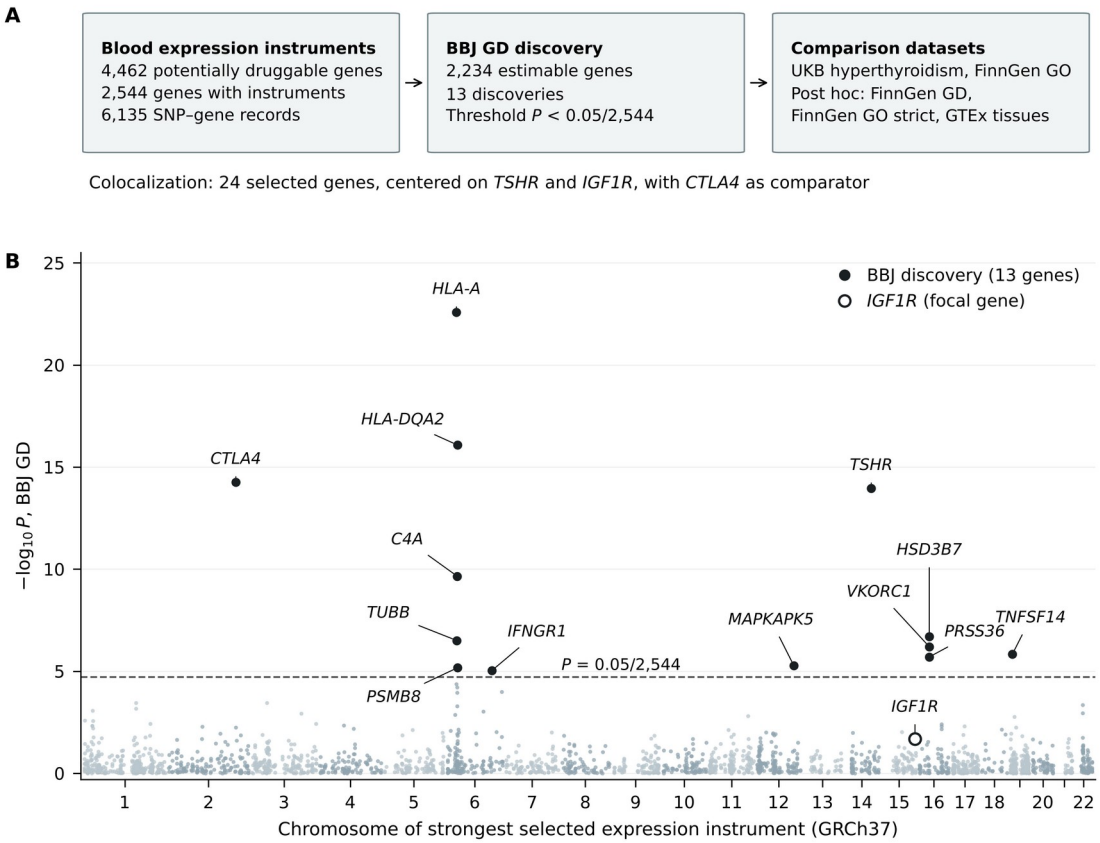


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Figure 2

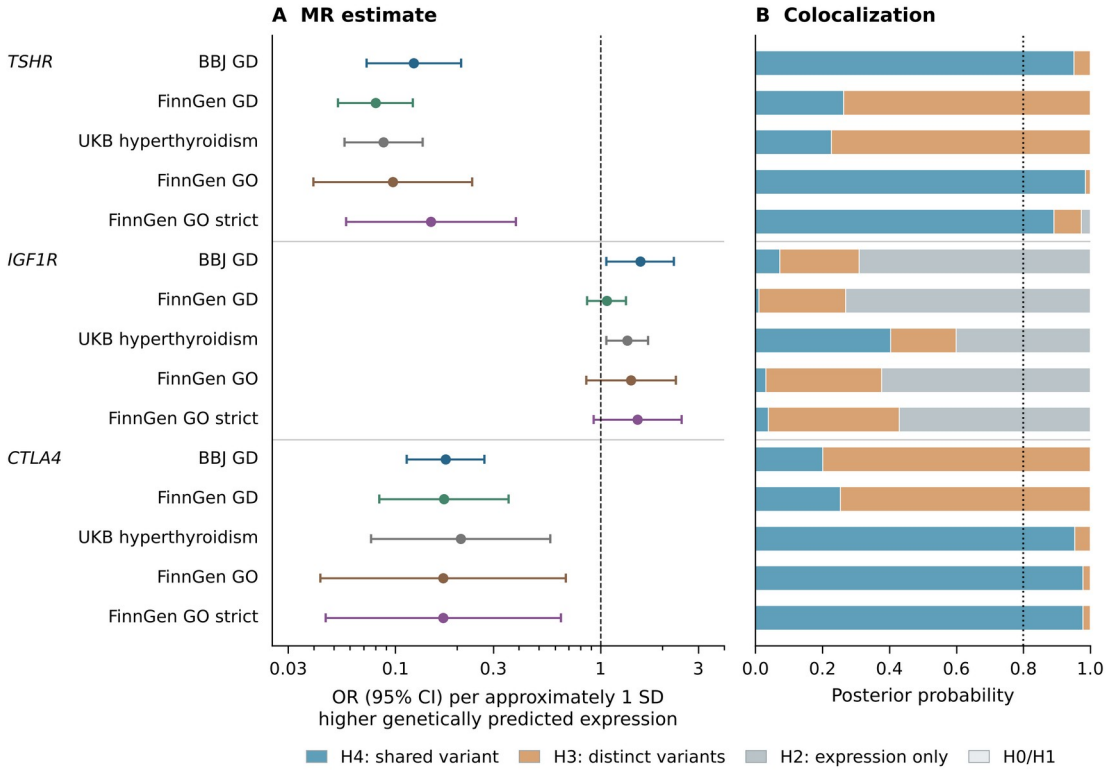


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Figure 3

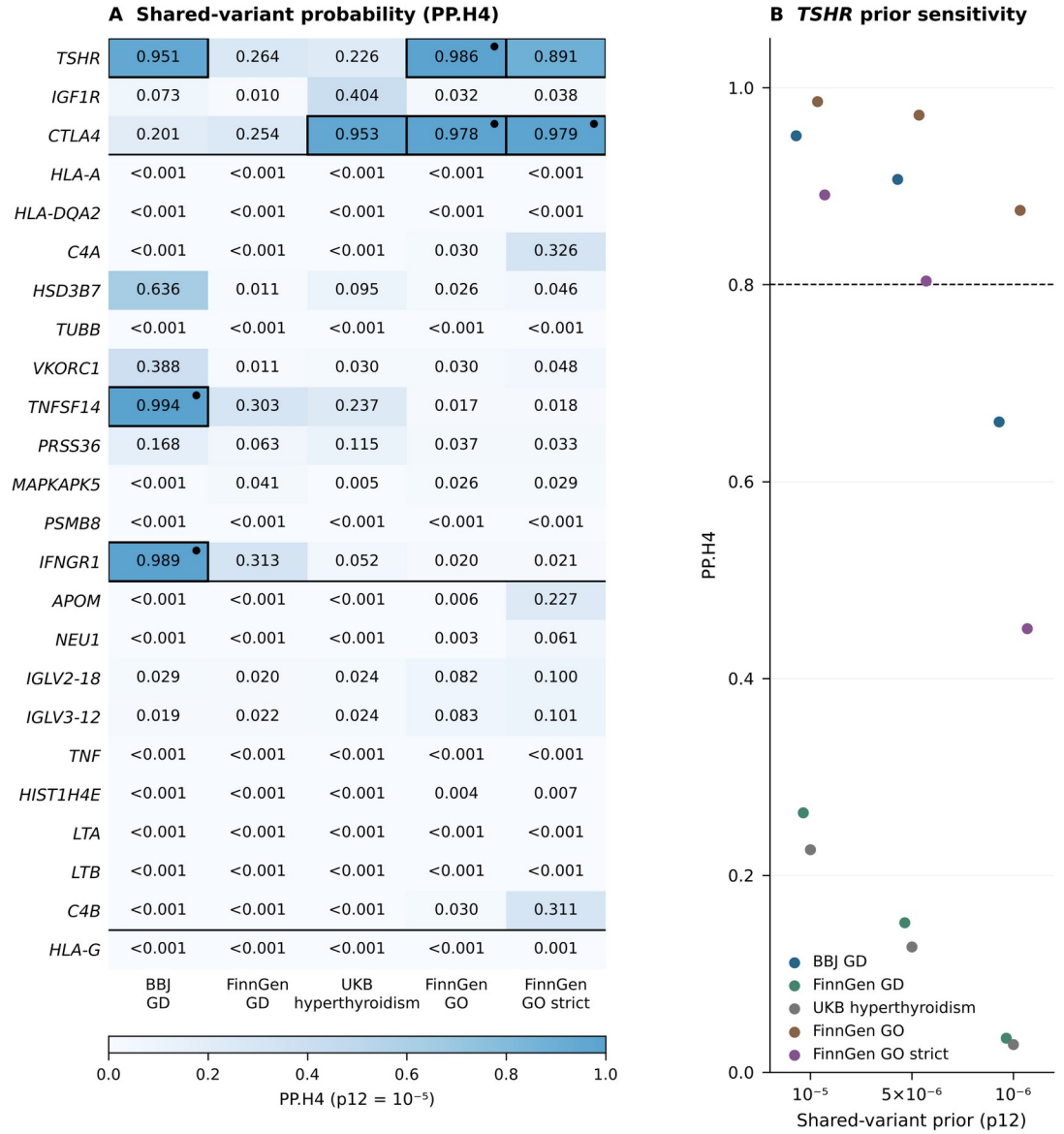


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Supplementary material

TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

Supplementary methods

Dataset definitions and analysis chronology

BBJ GD (GCST90018627) and UKB self-reported hyperthyroidism/thyrotoxicosis (GCST90038636) retained the source sample sizes and estimates. UKB used BOLT-LMM linear-model statistics; its report included adjustment for sex, age, BMI, assessment centre, ethnicity, genotyping batch and 20 principal components. The available documentation did not establish an exclusively European analysis sample. FinnGen R12 GWAS counts were 858/499,490 for GRAVES_OPHT, 3,962/496,386 for E4_GRAVES_STRICT and 753/499,595 for E4_GRAVES_OPHT_STRICT. Counts refer to the analyzed GWAS files, not later phenotype-browser totals. Strict GD combines specified GD/ophthalmopathy diagnostic criteria with the Graves condition and exclusion of amiodarone exposure; strict ophthalmopathy requires its specified ophthalmopathy and thyrotoxicosis codes. The exact endpoint definitions and source manifests are retained with the analysis verification records. None supplies GD-only controls.

The original three-outcome analysis preceded the two FinnGen extensions. The expanded analysis plan was recorded before the new endpoint associations were computed, but it was not a prospective registration of the complete study. The 24 colocalization genes comprise 13 BBJ discoveries, *IGF1R*, nine additional primary FinnGen GD discoveries, and frequency-sensitive *HLA-G*. No new discovery threshold was optimized from these results.

Instrument selection and effect reconstruction

Significant cis-eQTL instruments ($P < 5\times10^{-8}$ within ±1 Mb of the gene) were clumped at $r^2 < 0.001$ within 10 Mb using the European 1000 Genomes Phase 3 reference (n = 503). Selected instruments and GRCh37 positions are in the Instruments worksheet. $SE = 1/\sqrt{[2f(1-f)(N + Z^2)]}$ and $\beta = Z \times SE$, where f is the assessed-allele frequency and N is the variant-specific expression sample size. Unit expression variance was assumed. Variant $F = Z^2$. The estimate is approximately per expression SD, not per SD of the genetic predictor.

TwoSampleMR harmonise_data(action = 2) aligned alleles and excluded ambiguous palindromic variants with MAF > 0.42. Missing outcome estimates were not imputed. BBJ and FinnGen retained log-odds scales; UKB β and SE were each divided once by $\mu(1-\mu)$, with $\mu = 3{,}731/484{,}598$. GRCh37 expression and GRCh38 FinnGen coordinates were not directly joined: identifiers and allele pairs

were checked, with duplicate and alias audits. Source covariate adjustments were retained; patient-level re-adjustment and subgroup analysis were unavailable.

MR estimators and instrument diagnostics

For a single instrument, the Wald coefficient is β_Y/β_X and its first-order SE is $SE_Y/|\beta_X|$. Multiple-instrument MR used inverse-variance weighted regression through the origin. Multiplicative random-effects SEs included under-dispersion correction, so heterogeneity could widen the interval. Normal two-sided tests and 95% CIs were used for the primary estimators. The alternative-estimator worksheet preserves weighted median, weighted mode and MR-Egger results for the original three outcomes; mode and Egger intervals use the corresponding t distributions. These alternative estimators were not rerun for all new endpoints. With three or four *IGF1R* instruments, these estimators had little power.

Post hoc leave-one-out analyses applied IVW when at least two variants remained and the Wald ratio when one remained. Single-instrument *TSHR* and BBJ *CTLA4* were ineligible. Original three-outcome results, new-endpoint results for the fixed 14-gene set, and 72 *C4B*/*HLA-G* omission estimates under two frequency sources are retained in separately named worksheets. Independent calculations reproduced the eight full-set and 72 omission estimates for *C4B* and *HLA-G*. They reuse the same association data and are not independent replications. *C4B*/*HLA-G* instrument weights use β_X^2/SE_Y^2, normalized within each full model. A single-SNP *P* value can be much smaller than the full IVW *P* value when heterogeneity inflates the full-model SE.

TSHR clumping sensitivity used eQTL thresholds of $5\times10^{-8}/5\times10^{-6}$, r^2 thresholds of 0.001/0.01/0.1, and European (n = 503) or East Asian (n = 504) reference panels. Primary settings selected one European and two East Asian variants, and rs179252 was retained in every setting. Chromosome 16p11.2 analyses found high East Asian LD among rs4889606, rs34649473 and rs78924645; conditional analysis supported regional dependence rather than three independent gene effects.

TSHR allele alignment and reference LD

In a post hoc audit, original full eQTLGen and disease GWAS files were re-extracted for rs179252 and rs179247. Their GRCh37 positions were 14:81435985 and 14:81432546, and GRCh38 positions were 14:80969641 and 14:80966202. Both non-palindromic allele pairs matched across sources. UKB effects were reversed to the G allele after the existing scale conversion; BBJ and FinnGen required no reversal. Exposure coefficients and first-order Wald estimates were independently reconstructed. Original GWAS *P* values and recalculated MR *P* values were retained separately.

G-allele counts at both variants were positively correlated in European (n = 503; r = 0.978322812; r^2 = 0.957115524) and East Asian (n = 504; r = 0.995803919; r^2 = 0.991625445) references, without

missing genotypes. Direct BED decoding agreed with PLINK dosage output. PLINK haplotype estimates supported G–G and T–A coupling; these were inferred from unphased genotypes, not directly observed phased haplotypes. Genotype-based r^2 and haplotype-based estimates were kept distinct. Alignment to the G allele resolved the negative default European correlation obtained when counting T at rs179252. The TSHR_allele_audit worksheet retains both eQTL records, both variants across five disease outcomes, all four inferred haplotypes in each population, and MR reproduction. This audit does not identify a causal variant or establish mediation in thymus or orbit.

Colocalization hypotheses and regional inputs

coloc.abf received beta/varbeta data for disease (dataset 1, case-control type) and expression (dataset 2, quantitative type, sdY = 1). Outcome allele frequency was not used in the Bayes factors. Complete available cis-eQTL statistics, not just significant eQTLs, were matched by identifier and allele compatibility. H0 denotes neither trait associated; H1 disease only; H2 expression only; H3 both traits associated through distinct variants; H4 both traits associated through one shared variant. PP.Hk is the posterior probability of hypothesis k, conditional on the model and priors. Dataset order is essential to interpreting H1 and H2.

Primary priors were p1 = p2 = 10^{-4} and p12 = 10^{-5}; sensitivity p12 values were 5×10^{-6} and 10^{-6}. The PP.H4 ≥ 0.80 threshold was a reporting convention, not a significance test. H0–H4 sum to one apart from rounding. Values displayed as <0.001 are nonzero full-precision probabilities in the workbook. The top_snp is conditional on H4, not necessarily the disease lead variant. H3 is a competing distinct-variant model, not a count of causal signals. Multi-signal colocalization was not performed.

Four frequency/availability scenarios were used: original_reference (original available reference frequencies), paired_reference and paired_eqtlgen (the same variants with the respective frequencies), and available_eqtlgen (all matching variants with valid cohort frequencies). There are 1,440 gene–outcome–scenario–prior rows. The additional *TSHR* analysis comprises four windows × five outcomes × four scenarios × three priors = 240 rows. Windows were the full source cis-region, ±500 kb and ±250 kb around the recorded GRCh37 GenePos anchor, and the five-outcome variant intersection. The full cis-source contained 7,309 eQTL rows; all three FinnGen full-region analyses matched 6,579 variants, and the five-outcome intersection contained 4,030. Raw outcome regional scans checked whether the strongest disease variants survived matching. New candidate coverage checks also examined marginal expression strength; the absent rs5759410 lead in BBJ was checked against variant aliases without finding a replacement.

For *TSHR*, *IGF1R* and *CTLA4*, colocalization and MR in FinnGen GD and strict ophthalmopathy were independently reproduced from re-downloaded FinnGen R12 regional statistics using two separately written implementations, with the original ophthalmopathy endpoint as a control. The

allele-compatible variant sets were the same (6,675, 7,850 and 4,993 variants), and the top variant was the same in all 27 gene–endpoint–prior settings, including the control. The reproduction used gnomAD v2.1.1 European frequencies, which were available for more of these variants than the 1000 Genomes European frequencies used in the reported analysis (6,674, 7,850 and 4,992 analysed, versus 6,579, 7,740 and 4,924). On the control endpoint, PP.H4 changed by at most 4.1×10^{-4}; for *TSHR* this difference came from the additional variants. Reported new-endpoint PP.H4 values were reproduced within 1.2×10^{-3}, and no setting changed side of 0.80. The *TSHR* strict-ophthalmopathy value at p12 = 10^{-6} was 0.449 rather than 0.451 (0.451 on the reported variant set), and *IGF1R* PP.H2 and PP.H3 differed by up to 0.003, as on the control endpoint. MR estimates were reproduced to numerical precision on the original exposure scale.

Frequency substitution and precision benchmarks

The published eQTLGen frequency release covered 26,609 participants, excluding Framingham. Reconstructing β_X and SE_X with fixed Z and N preserves their ratio up to floating-point precision. For a single-IV Wald estimate, frequency scaling changes the effect unit and SE together and preserves the normal-test *P* value; for IVW, variant weights can change. The new-endpoint MR dataset separates identical-variant scaling from repeat allele harmonization. *C4B* and *HLA-G* were audited with identical retained variants and outcome effects, so their differences arise from scaling and weighting rather than a changed variant set. Large frequency discrepancies are documented, not assumed to be resolved by correct identifier matching.

The detection benchmarks use $(z_{1-\alpha/2} + z_{0.80}) \times SE$ as the approximate 80% minimum detectable absolute log-OR. Alpha was 0.05/2,544 for BBJ and 0.05 for the original UKB/FinnGen comparisons. These are precision summaries using observed SEs, not prospective sample-size calculations, post hoc proof of power, or power of colocalization/the combined filter. The Detection_original worksheet retains these three-outcome results. Applying the same formula to the MR_primary standard errors for FinnGen GD at alpha 0.05/2,544 gave 24.8% and 52.2% of 2,480 estimable genes with 80% power for ORs of 1.5 and 2.0. At alpha 0.05, the detectable *IGF1R* ORs were 1.72 in BBJ, 1.39 in UKB and 1.36 in FinnGen GD.

Fixed-panel tissue extension

The post hoc tissue extension retained a frozen panel of *TSHR* and *IGF1R* in five standard-release GTEx v8 tissues. QTL sample counts were 574 (thyroid), 483 (cultured fibroblasts), 581 (subcutaneous adipose), 469 (visceral adipose) and 670 (whole blood). The original v8 archive supplied significant variant–gene pairs and eGenes records for all tested genes; gene-specific release thresholds exceeded 5×10^{-8} for every eligible combination, permitting complete selection at the fixed threshold. Absence from the tested-gene file was distinguished from a minimum eQTL *P* value above

the threshold. The standard release includes multiple ancestries; the overall donor cohort was 85.3% European American, not a tissue-specific percentage.

Variants were restricted to ±1 Mb of the source GENCODE v26 TSS, $P < 5\times10^{-8}$, F > 10 and MAF ≥ 0.01. GTEx variant IDs identify GRCh38 REF and ALT, with slope referring to ALT. Actual effect-allele frequencies were reconstructed from each selected eGene MAF and ref_factor, not by treating MAF as ALT frequency. GTEx mappings linked GRCh38 variants to rsIDs and GRCh37 reference coordinates. PLINK clumping used 503 European reference samples, $r^2 < 0.001$ and a 10-Mb window; an independent genotype-based calculation reproduced selection. One of 20 thyroid *IGF1R* candidates was absent from the reference, while its strongest candidate was retained. All selected variants were non-palindromic and matched exactly in the five disease datasets; no proxies or relaxed thresholds were used.

Each of the four estimable gene–tissue combinations retained one SNP. Wald estimates used β_Y/β_X and $SE_Y/|\beta_X|$ with normal two-sided *P* values and 95% CIs; independent calculations reproduced all 20 estimates. The UKB linear-to-log-odds conversion was applied once. ORs represent one unit of inverse-normal-transformed normalized expression; they are not directly comparable to an intervention or to the reconstructed eQTLGen scale. Primary adjusted $P = \min(1, 10P)$ for FinnGen GD; secondary adjusted $P = \min(1, 40P)$ for the other datasets. The 30 unestimable cells remained in the planned families. Single-instrument analyses cannot support heterogeneity, leave-one-out or MR-Egger diagnostics. Significant-only eQTL pairs were not used for colocalization; full regional data were not obtained from the requester-pays resource, and no tissue posterior was calculated. Tissue_extension in the single Supplementary Dataset retains all 50 cells, availability, selected and excluded candidates, harmonized alleles and source statistics.

Orbital expression and reproducibility

Orbital adipose tissue was collected during orbital decompression from four patients with inactive TED; the control was nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Paired-end RNA sequencing (Illumina NovaSeq) used two technical-replicate libraries per donor, which were combined by biological sample; reads were aligned to GRCh38 and quantified as TPM. Figure S1 shows sample-level transcript abundance and $\log_2[(\text{mean TED TPM} + 0.01)/(\text{control TPM} + 0.01)]$. One control cannot estimate control-group biological variance reliably, and no differential-expression *P* value was used. The dataset is descriptive and did not influence candidate selection.

The workbook Readme and Column_dictionary sheets describe every sheet; reproduction validates implementation, not causal assumptions.

Table S1. Additional FinnGen GD screen discoveries

Gene	Chr	FinnGen GD IVs	FinnGen GD OR (95% CI)	FinnGen GD *P*	BBJ PP.H4	FinnGen GD PP.H4	FinnGen GO strict PP.H4
APOM	6 (MHC)	1	0.05 (0.04–0.06)	1.43×10^{-108}	<0.001	<0.001	0.227
NEU1	6 (MHC)	1	0.04 (0.03–0.06)	1.43×10^{-108}	<0.001	<0.001	0.061
IGLV2-18	22	1	0.56 (0.48–0.66)	7.52×10^{-12}	0.029	0.020	0.100
IGLV3-12	22	1	1.20 (1.14–1.26)	7.52×10^{-12}	0.019	0.022	0.101
TNF	6 (MHC)	1	0.60 (0.51–0.71)	4.65×10^{-10}	<0.001	<0.001	<0.001
HIST1H4E	6 (MHC)	1	2.54 (1.88–3.43)	1.51×10^{-9}	<0.001	<0.001	0.007
LTA	6 (MHC)	1	0.46 (0.35–0.61)	3.75×10^{-8}	<0.001	<0.001	<0.001
LTB	6 (MHC)	1	0.18 (0.10–0.36)	4.89×10^{-7}	<0.001	<0.001	<0.001
C4B	6 (MHC)	2	1.63 (1.31–2.03)	1.10×10^{-5}	<0.001	<0.001	0.311

Genes outside the BBJ discovery set that passed $P < 0.05/2{,}544$ in the post hoc full FinnGen GD screen (1000 Genomes European reference frequencies). These are not independent of the screen and seven of nine are in the MHC. *APOM*/*NEU1* share rs3101018 and *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. None reached PP.H4 ≥0.80 ($p12 = 10^{-5}$) in any dataset. Chr, chromosome; CI, confidence interval; IV, instrumental variable; MHC, major histocompatibility complex; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S2. Frequency-sensitive exploratory candidates

Gene	Dataset	Frequency source	IVs	OR (95% CI)	*P* value	Q-test *P*
C4B	FinnGen GD	1000G EUR	2	1.63 (1.31–2.03)	1.10×10^{-5}	7.79×10^{-7}
C4B	FinnGen GD	eQTLGen cohort	2	1.75 (1.31–2.33)	1.48×10^{-4}	1.36×10^{-8}
C4B	FinnGen GO strict	1000G EUR	2	1.68 (1.44–1.95)	2.74×10^{-11}	0.122
C4B	FinnGen GO strict	eQTLGen cohort	2	1.80 (1.46–2.23)	6.17×10^{-8}	0.0587
HLA-G	FinnGen GD	1000G EUR	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G	FinnGen GD	eQTLGen cohort	16	0.60 (0.47–0.75)	1.72×10^{-5}	2.26×10^{-10}
HLA-G	FinnGen GO strict	1000G EUR	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G	FinnGen GO strict	eQTLGen cohort	16	0.70 (0.53–0.92)	0.0122	0.131

Exposure effects were reconstructed with 1000 Genomes Phase 3 European reference allele frequencies (1000G EUR; primary) or eQTLGen cohort frequencies; variant sets and outcome effects were held fixed, so changes reflect rescaled expression effects and IVW weights. *HLA-G* was added because its eQTLGen-frequency *P* crossed the screen threshold ($P < 0.05/2{,}544$). Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are in the workbook. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S3. *TSHR* regional colocalization and prior sensitivity

Dataset	Matched variants	Disease lead (*P*)	rs179252 *P*	rs179252-G OR (95% CI)	H4 top variant	PP.H4, 10^{-5} / 10^{-6}	PP.H4 range across windows
BBJ GD	4,336	rs28414437 (1.83×10^{-15})	1.09×10^{-14}	0.80 (0.76–0.85)	rs179252	**0.951** / 0.661	0.951–0.951
FinnGen GD	6,579	rs1023586 (1.58×10^{-36})	3.19×10^{-32}	0.77 (0.73–0.80)	rs1023586	0.264 / 0.035	0.264–0.264
UKB hyperthyroidism	5,667	rs4903961 (1.17×10^{-43})	8.77×10^{-28}	0.77 (0.74–0.81)	rs1023586	0.226 / 0.028	0.226–0.226
FinnGen GO	6,579	rs3783947 (1.23×10^{-7})	2.82×10^{-7}	0.78 (0.71–0.86)	rs179252	**0.986** / **0.875**	0.986–0.987
FinnGen GO strict	6,579	rs2110695 (1.34×10^{-5})	9.02×10^{-5}	0.82 (0.74–0.90)	rs179252	**0.891** / 0.451	0.891–0.905

Disease lead, the strongest disease-associated variant among the matched cis variants, with its *P* value; rs179252 *P*, z-test *P* of the aligned G-allele disease association (the UKB source BOLT-LMM *P* was 1.3×10^{-27}). H4 top variant, the variant with the highest posterior under the shared-variant hypothesis; it is not necessarily the disease lead. PP.H4 under p12 = 10^{-5} and 10^{-6} (p1 = p2 = 10^{-4}); bold, ≥0.80. Window range covers the full source cis-region, ±500 kb, ±250 kb and the five-outcome intersection at p12 = 10^{-5}; it is not a confidence interval. All 240 window settings are in TSHR_windows. BBJ pairs European eQTL with East Asian disease statistics. CI, confidence interval; eQTL, expression quantitative trait locus; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S4. *TSHR* and *IGF1R* tissue expression extension

A. Availability

Gene / tissue	GTEx N	Instrument (F)	Status
TSHR / Thyroid	574	—	No instrument (minimum eQTL $P = 9.60\times10^{-8}$)
IGF1R / Thyroid	574	rs1815009 (F = 61.1)	Estimable
TSHR / Cultured fibroblasts	483	—	Not tested by GTEx in this tissue
IGF1R / Cultured fibroblasts	483	rs6598538 (F = 33.8)	Estimable
TSHR / Subcutaneous adipose	581	rs12101255 (F = 38.5)	Estimable
IGF1R / Subcutaneous adipose	581	—	No instrument (minimum eQTL $P = 3.56\times10^{-7}$)
TSHR / Visceral adipose	469	rs74064796 (F = 32.1)	Estimable
IGF1R / Visceral adipose	469	—	No instrument (minimum eQTL $P = 7.76\times10^{-6}$)
TSHR / Whole blood	670	—	No instrument (minimum eQTL $P = 2.46\times10^{-4}$)
IGF1R / Whole blood	670	—	No instrument (minimum eQTL $P = 5.42\times10^{-5}$)

B. Estimates

Gene / tissue	Dataset	Family	OR (95% CI)	*P* value	Below family threshold	Direction vs blood
TSHR / Subcutaneous adipose	BBJ GD	Secondary	0.46 (0.38–0.55)	1.92×10^{-15}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GD	Primary	0.36 (0.31–0.43)	1.28×10^{-34}	Yes	Same
TSHR / Subcutaneous adipose	UKB hyperthyroidism	Secondary	0.31 (0.27–0.37)	2.86×10^{-42}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO	Secondary	0.45 (0.32–0.63)	5.76×10^{-6}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO strict	Secondary	0.51 (0.35–0.74)	3.67×10^{-4}	Yes	Same
TSHR / Visceral adipose	BBJ GD	Secondary	0.56 (0.46–0.67)	2.64×10^{-10}	Yes	Same
TSHR / Visceral adipose	FinnGen GD	Primary	0.50 (0.43–0.58)	4.09×10^{-19}	Yes	Same
TSHR / Visceral adipose	UKB hyperthyroidism	Secondary	0.39 (0.34–0.45)	9.19×10^{-38}	Yes	Same
TSHR / Visceral adipose	FinnGen GO	Secondary	0.69 (0.49–0.98)	0.0368	No	Same
TSHR / Visceral adipose	FinnGen GO strict	Secondary	0.78 (0.54–1.13)	0.186	No	Same
IGF1R / Thyroid	BBJ GD	Secondary	1.23 (0.90–1.68)	0.200	No	Same
IGF1R / Thyroid	FinnGen GD	Primary	0.76 (0.56–1.02)	0.0692	No	Opposite
IGF1R / Thyroid	UKB hyperthyroidism	Secondary	1.33 (0.99–1.80)	0.0594	No	Same
IGF1R / Thyroid	FinnGen GO	Secondary	1.02 (0.54–1.94)	0.945	No	Same
IGF1R / Thyroid	FinnGen GO strict	Secondary	1.23 (0.62–2.44)	0.549	No	Same
IGF1R / Cultured fibroblasts	BBJ GD	Secondary	0.91 (0.70–1.20)	0.511	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GD	Primary	1.11 (0.89–1.38)	0.359	No	Same
IGF1R / Cultured fibroblasts	UKB hyperthyroidism	Secondary	1.03 (0.82–1.29)	0.815	No	Same
IGF1R / Cultured fibroblasts	FinnGen GO	Secondary	0.78 (0.49–1.25)	0.301	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GO strict	Secondary	0.78 (0.47–1.30)	0.341	No	Opposite

Post hoc fixed-panel extension using GTEx v8; tissue colocalization was not performed. A, availability of the ten gene–tissue combinations; F statistics are for the single retained instrument. ‘Not tested’ means the gene is not among the genes GTEx tested in that tissue. GTEx whole blood (n = 670) yielded no instrument at the fixed threshold, whereas the main analysis used eQTLGen (up to 31,684 participants). B, all 20 estimates. The primary family is the ten FinnGen GD comparisons ($P < 0.005$); the secondary family is the 40 comparisons with the other datasets ($P < 0.00125$); unestimable combinations stay in both denominators. ORs are per unit of inverse-normal-transformed tissue expression and are not comparable in size with the blood ORs. With one instrument, each *P* value equals that variant’s disease-association z-test *P* value (UKB source BOLT-LMM *P* values differ slightly). The adipose *TSHR* instruments (rs12101255, rs74064796; 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and re-express the regional association rather than providing independent evidence. Cultured fibroblasts and adipose tissue are not orbital tissue. Direction, compared with the blood estimate in the same dataset. CI, confidence interval; eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure S1

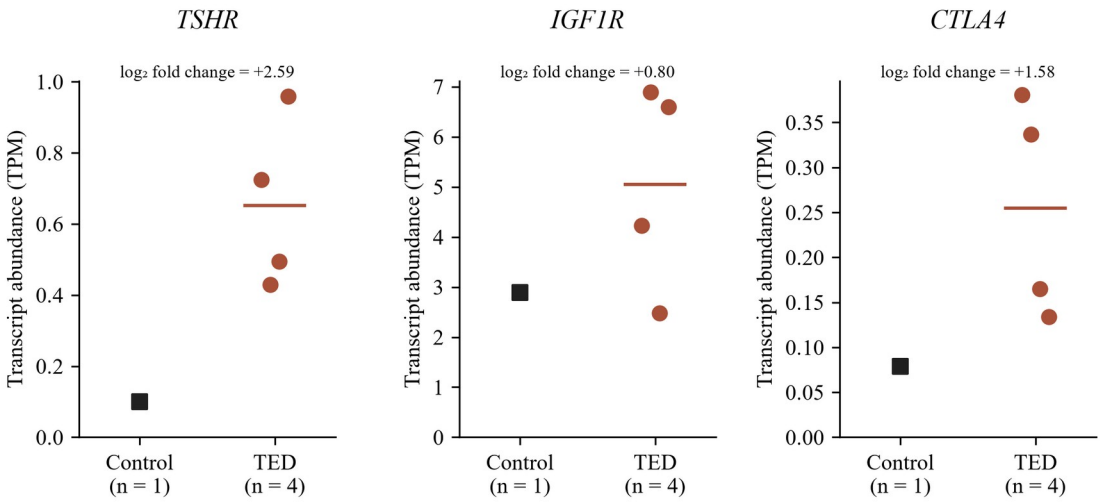


Figure S1. Descriptive orbital expression of *TSHR*, *IGF1R* and *CTLA4*. Transcript abundance in orbital adipose tissue collected during orbital decompression from four patients with inactive TED, and in nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Points are biological samples (technical replicates combined); horizontal lines show TED means. The $\log_2$ fold change is $\log_2$[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Vertical axes differ by gene. The control came from a different fat depot, and one control cannot estimate control-group variance; no inferential test was performed, and these observations do not validate the genetic findings. TED, thyroid eye disease; TPM, transcripts per million.

Figure S2

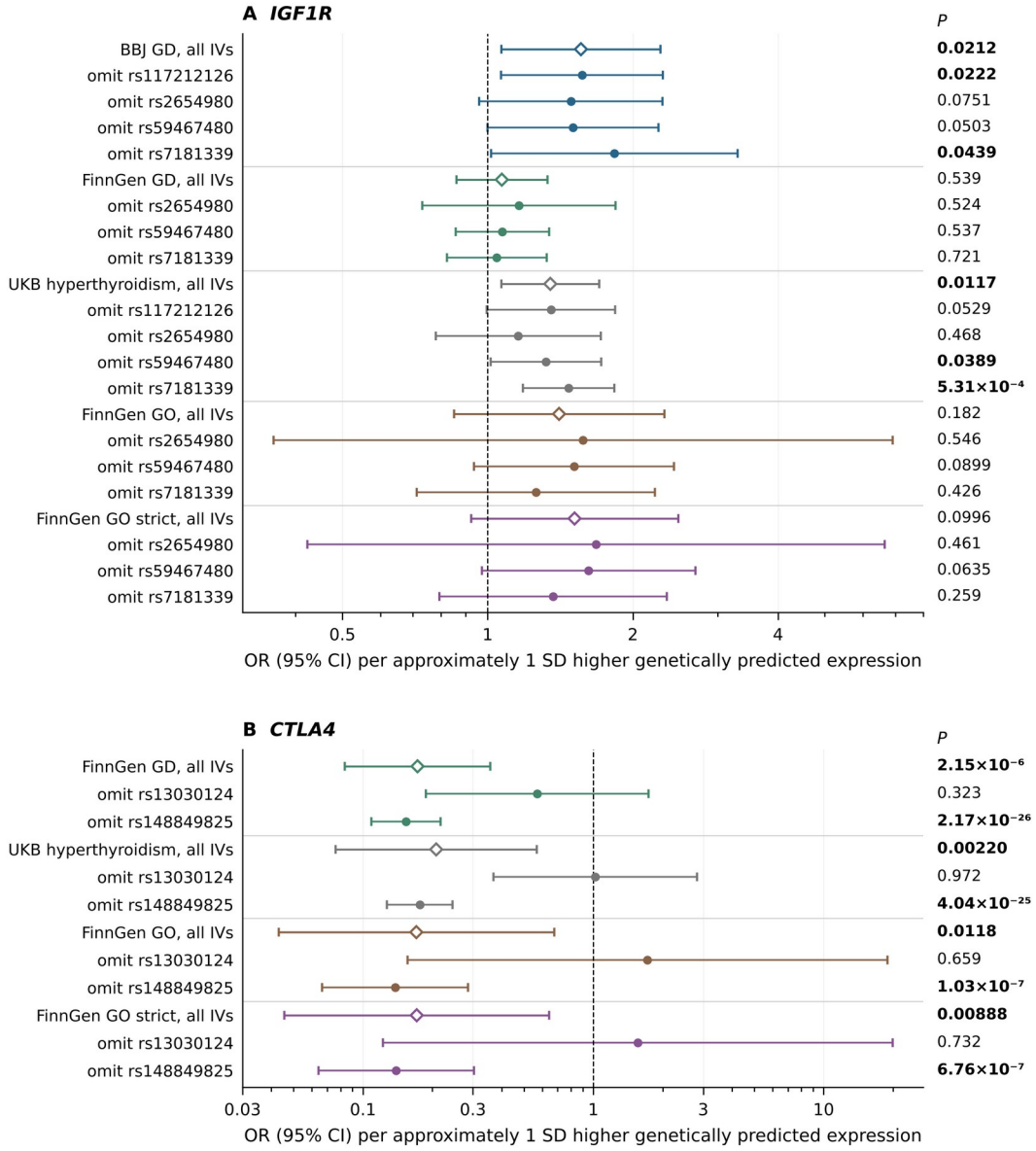


Figure S2. Dependence of *IGF1R* and *CTLA4* estimates on individual instruments. For each dataset, the open diamond is the estimate using all instruments (as in Table 2), and the filled points below it omit the named instrument (IVW when at least two instruments remain, Wald ratio when one remains). Lines are 95% CIs on a logarithmic axis; the dashed line is OR = 1. *P* values are shown at right; bold, nominal $P < 0.05$. (A) *IGF1R* in all five datasets. (B) *CTLA4* in the four FinnGen and UKB datasets; BBJ *CTLA4* and all *TSHR* estimates use one instrument and could not be assessed. The 25 post hoc omission estimates reuse the same association data and are not independent replications; loss of nominal significance can reflect reduced precision as well as the omitted variant's contribution. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.